3CC3 Design Project Proposal

Project Weight: 15 Marks
Project Due Dates:

Deliverables	Monday Lab Sessions
Project Idea (s) (1 Paragraph)	2nd -November-25 (End of day)
Project Proposal	9th -Nov-21 (Prior to lab start)
Demo and Video Submission	24th -November-25

Project Execution Timeline
1. One paragraph that highlights your project idea will be submitted according to the due dates in the Table above.
2. During your lab session on Nov 3rd, the Lab instructor will spend 5-10 min discussing the lab project idea with you. The instructor can accept, modify, or reject and suggest another project idea to the student.
3. The project Proposal will be submitted on Nov 9th. This document is based on the prior project idea discussion with the lab instructor. The instructor will discuss with you the details of the proposal.
4. The project execution will start on Nov 9th, and it should be completed within three weeks. The project demos will be conducted during your lab session on Nov 24th.
5. Lab attendance is required during the three weeks of the project. Progress will be monitored and assessed by the lab instructor.
6. Since the project hardware is available to every student and it is expected to complete it at home. The lab view programing should be reserved for the lab time. If a student needs more time for lab view, they can consider buying the student version.

Project Goal:
The goal of this project is to design a basic **smart system** utilizing your knowledge and experience from the 3CC3 labs. The project must include sensor(s), actuator(s), cloud, HMI(s).

Project Components:
1. You may use any of the components that are provided to you in the **Mac IoT & Controls kit**.
2. You may also use some components that you purchase on your own (you must consult with your lab instructor and let him/her know about your plans before purchasing any component).

Project Individual or Team-based execution
The project should be performed as team formed by the lab groups.

Project Requirements:

1. Complexity	The smart system needs to have a degree of complexity equivalent to **3 combined** labs.
	A proposal with too little complexity/workload or too much complexity/workload may not be approved. However, a tougher project has a potential to generate a higher mark, versus a simpler project.
2. Hardware	Your project must contain **at least 2 sensors and 1 actuator**.
	The maximum number of sensors/actuators is 6 (unless approved by the lab instructor).
3. Software	• Your application must contain sufficiently complex Logic and Programming. • You need to create a **Local HMI** for your system. **LabVIEW** is highly recommended to create your HMI. If you choose a software **other than LabVIEW**, your HMI quality should be similar or better than the one of LabVIEW. • You need to create a **web-based HMI** for your system.
4. Purpose	Your smart system must have a clear purpose related to a real-world application.
5. Low cost and small-scale prototype	Try to build a small scale and a low-cost prototype for the application. You can use cheap materials like *cardboard or wood*. Be creative! For example, if you are monitoring the temperature in multiple rooms in a building, then build a little mock-up with multiple rooms.

Examples of Projects:

1. Design an **elevator system** – using limit switches and a motor
2. Design a **home security system** – using motion sensors
3. Design a **temperature monitoring and control** system for a building.
5. Design a **sun tracking system** for a solar panel.

See sample videos of previous projects that are posted on avenue to learn.

Role of the Lab Instructor

1. Lab instructors will review your project idea, approves it, modify, or reject and suggest another idea.
2. Troubleshooting your program and any hardware issues is your responsibility, not the responsibility of the lab instructor.
3. Lab instructors will monitor your progress every week. At the end of every lab session, you **must** discuss with your lab instructor the progress that you made in the project.

When can you work on your lab project?

Since you have the components at home, you can work on your project at any convenient time in addition to the lab time.

Deliverables

Deliverable	**Deadline**	**Requirement**
1. Project Idea (s)	02-Nov-25 (End of day)	- **Only 1 paragraph**
2. Project Proposal	09-Nov-25 (Prior to lab start)	- **2-3 pages** - abstract - description of project, its purpose and functionality - a List of all devices, sensors, actuators, electronic components that are required to complete your project. - Electrical schematic drawing showing the connections of the project components. You may use https://easyeda.com/ for that purpose. - A time-line detailing the major goals for each week.
3. real life demo of the project	24-Nov-25	- all work in the projects will stop for the last hour of the lab time. Only DEMOs will be running during the last hour.
4. Video and Code submission	24-Nov-25	- Prepare a 3-4 min video describing your project hardware, software and demonstrating its main functionalities.

Project Marking Scheme

Project idea (1 paragraph)	0.5 Marks	Clear description of the project goal and functions
Proposal & Schedule	1 Marks	Clear diagrams - proper description of the operation – clear timeline
Project Functionality	3 Marks	All components discussed in the proposal are fully functional
DEMO Presentation	2.5 Marks	Clear presentation and explanation of the project
Video	2.5 Marks	Clear video demo that shows the functionality of the project
Complexity	4 Marks	Local HMI - Arduino - number of sensors and actuators – Logic and code complexity – Web-based HMI - Cloud features (database, alerts, logic, etc)
Lab attendance and progress	1.5 Marks	Lab Attendance – weekly progress is matching the plan in the project proposal
Total	**12.5** Marks	